

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

1st Semester of B. Pharm. Examination

University Theory Examination December 2016

PH112 - HUMAN ANATOMY & PHYSIOLOGY-I

Date: 26.12.2016, Monday

Time: 10:00a.m. to 1:00p.m.

Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.

SECTION - I

- Q 1** Attempt all questions. Each question is of one mark. **20**
1. Blood is a complex _____ tissue and is the only _____ tissue in the body.
[A] connective, fluid
[B] connective, colloid
[C] nervous, saline
[D] muscle, fluid
 2. The living cells and cell fragments in blood are called the _____ elements and include erythrocytes, leukocytes, and _____.
[A] buffy coat, plasma
[B] hematocrit, neutrophil
[C] formed, platelets
[D] plasma, platelets
 3. The molecule most responsible for carrying oxygen in blood is known as _____.
[A] hemoglobin
[B] monocytes
[C] globulin
[D] albumin
 4. If a person has type AB blood, then that person can receive blood from which type donors?
[A] people with type AB but not O, A, or B
[B] people with types A, B, or O
[C] only people with types AB or O
[D] people with types AB, A, B, or O
 5. The tiny mass of cells that forms the pacemaker of the heart is called the _____.
[A] bundle of his
[B] purkinje fibre
[C] SA node
[D] AV node

6. The cardiovascular and _____ systems work together to supply the body with oxygen and rid it of carbon dioxide.
[A] Urinary
[B] Digestive
[C] Nervous
[D] Respiratory
7. Which of the following is not an organ of the respiratory system?
[A] Bronchi
[B] Thymus
[C] Lungs
[D] Larynx
8. Rings of _____ cartilage reinforce the trachea.
[A] articular
[B] hyaline
[C] epiphyseal
[D] fractured
9. The left lung has _____ lobes and the right lung has _____.
[A] 2, 3
[B] 1, 4
[C] 3, 2
[D] 1, 2
10. The _____ capacity is the total amount of exchangeable air in the lungs.
[A] vital
[B] residual
[C] tidal
[D] expiratory
11. What term means the ability of the body to maintain a stable internal environment despite changes in the external environment?
[A] static control
[B] homeostasis
[C] stimuli
[D] histology
12. You would look for the mediastinum in which cavity?
[A] spinal
[B] buccal
[C] nasal
[D] thoracic
13. All of the following are organs of the urinary system **except** the _____.
[A] adrenals
[B] kidney
[C] urethra
[D] urinary bladder
14. The nephron is responsible for _____ formation.
[A] fecal
[B] mucus
[C] saliva
[D] urine
15. The color of normal urine is attributed to a pigment resulting from the destruction of _____.
[A] hemoglobin
[B] sugar
[C] fat
[D] bile
16. The small intestine joins the large intestine at the _____ valve.

- [A] ileocecal
[B] pyloric
[C] precapillary
[D] cardioesophageal
17. Freshly oxygenated blood enters the heart through the _____, and is pumped out the _____.
[A] Right atrium; aorta
[B] left atrium; aorta
[C] right ventricle; pulmonary arteries
[D] right ventricle; aorta
18. Cardiac output can be determined by the following formula:
[A] Heart rate $\times$ stroke volume
[B] Heart rate – Stroke volume
[C] Heart rate + Stroke volume
[D] Heart rate / Stroke volume
19. What does the electrocardiogram (ECG) QRS wave indicate is happening in the heart?
[A] Atrial depolarisation
[B] Ventricular depolarisation
[C] Atrial repolarisation
[D] Ventricular repolarisation
20. Red blood cells survive around _____ days
[A] 100
[B] 110
[C] 120
[D] 130

SECTION – II

- Q 2** Attempt any **TWO** of the following
- A** Explain the anatomy of the heart with labeled schematic diagram. **05**
- B** Enlist the basic steps of respiration. Explain physiology of respiration. **05**
- C** Write a note on blood clotting. **05**
- Q 3** Attempt any **TWO** of the following
- A** Define blood pressure. Explain hormonal regulation of blood pressure. **05**
- B** Draw a labeled diagram of nephron. Explain the physiology of glomerular filtration. **05**
- C** Draw neat and well labeled diagram of digestive system and give the functions of liver. **05**

SECTION – III

- Q 4** Attempt any **FOUR** of the following
- A** Write a note on cardiac cycle. **05**
- B** Explain functions of blood and their components. **05**
- C** Define the following terminology **05**

05	1. Peptic ulcer disease	
05	2. Angina Pectoris	
05	3. Asthma	
05	4. Kidney stones	
05	5. Leukopenia	
05	Explain the phases of gastric acid secretion.	
05	Define homeostasis. Explain feedback mechanisms using suitable examples.	
05	Write a note on lung volumes and capacities.	
Q 5	Attempt any FOUR of the following	
05	Write a note on salivary glands.	
05	Explain cardiac output in detail.	
05	Explain the functions of White blood cells.	
05	Write a note on ECG.	
05	Explain life cycle of RBCs in detail.	
05	Discuss cardiac action potential.	